

CONVEX OPTIMIZATION

HOMEWORK 1

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1 (16 pts) – Convex sets

1.1 (12 pts) Closed sets and convex sets

1.1.1 Show that a polyhedron $\{x \in \mathbb{R}^n : Ax \leq b\}$, for some $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$, is both convex and closed.

Answer

Let $S = \{x \in \mathbb{R}^n : Ax \leq b\}$ with $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$.

Convexity. Take any $x, y \in S$ and $t \in [0, 1]$. By definition, $Ax \leq b$ and $Ay \leq b$. Since $t \geq 0$ and $1 - t \geq 0$, we may multiply the inequalities by these scalars without reversing the inequality directions:

$$A(tx) \leq tb, \quad A((1-t)y) \leq (1-t)b.$$

Adding these componentwise gives

$$A(tx + (1-t)y) \leq tb + (1-t)b = b.$$

Thus $tx + (1-t)y \in S$, which proves S is convex.

Closedness. Let $(x_k)_{k \geq 1} \subseteq S$ be a sequence that converges to some $\bar{x} \in \mathbb{R}^n$. Because the linear map $x \mapsto Ax$ is continuous (every linear operator on a finite-dimensional space is continuous), we have

$$\lim_{k \rightarrow \infty} Ax_k = A\bar{x}.$$

For each k , $Ax_k \leq b$. A convergent sequence of real vectors preserves componentwise inequalities; hence the limit satisfies $A\bar{x} \leq b$. Therefore $\bar{x} \in S$, so S contains all its limit points and is closed.

1.1.2 Show that if $S_i \subseteq \mathbb{R}^n, i \in I$ is a collection of convex sets, then their intersection $\bigcap_{i \in I} S_i$ is also convex. Show that the same statement holds if we replace “convex” with “closed”.

Answer

Let $\{S_i\}_{i \in I}$ be a family of subsets of $\mathbb{R}^n$.

Convexity. Assume each S_i is convex. Let $x, y \in \bigcap_{i \in I} S_i$ and $t \in [0, 1]$. For any $i \in I$, we have $x \in S_i$ and $y \in S_i$. Since S_i is convex, $tx + (1-t)y \in S_i$. This holds for every i , so $tx + (1-t)y \in \bigcap_{i \in I} S_i$. Hence the intersection is convex.

Closedness. Assume each S_i is closed. Let $(x_k) \subseteq \bigcap_{i \in I} S_i$ be a sequence that converges to $\bar{x} \in \mathbb{R}^n$. For any fixed $i \in I$, (x_k) is a sequence in S_i (because $\bigcap_{i \in I} S_i \subseteq S_i$). Since S_i is closed, the limit $\bar{x}$ belongs to S_i . As i was arbitrary, $\bar{x} \in \bigcap_{i \in I} S_i$. Thus the intersection is closed.

1.1.3 Given an example of a closed set in $\mathbb{R}^2$ whose convex hull is not closed.

Answer

Define

$$S = \{(x, 0) \mid x \in \mathbb{R}\} \cup \{(0, 1)\}.$$

That is, S consists of the entire x -axis together with the single point $(0, 1)$.

S is closed. The x -axis is closed because it is the zero level set of the continuous function $f(x, y) = y$. The singleton $\{(0, 1)\}$ is closed. A finite union of closed sets is closed; hence S is closed.

Convex hull of S . The convex hull $\text{conv}(S)$ consists of all convex combinations of points in S . Any point in $\text{conv}(S)$ either comes from two points on the x -axis (giving a point on the x -axis) or from $(0, 1)$ and a point $(a, 0)$ on the x -axis:

$$p = (1 - t)(a, 0) + t(0, 1) = ((1 - t)a, t), \quad t \in [0, 1].$$

For $t \in [0, 1)$ and arbitrary $a \in \mathbb{R}$, we obtain every point (x, y) with $0 \leq y < 1$ and $x \in \mathbb{R}$. For $t = 1$ we get exactly $(0, 1)$. Therefore

$$\text{conv}(S) = \{(x, y) \mid 0 \leq y < 1\} \cup \{(0, 1)\}.$$

$\text{conv}(S)$ is not closed. Consider the sequence $p_k = (1, 1 - \frac{1}{k})$ for $k = 1, 2, \dots$. Each p_k lies in $\text{conv}(S)$ because its second coordinate is < 1 . The limit is $\bar{p} = \lim_{k \rightarrow \infty} p_k = (1, 1)$. However $\bar{p} \notin \text{conv}(S)$: points with second coordinate 1 are in the convex hull only if they equal $(0, 1)$. Hence $\text{conv}(S)$ fails to contain one of its limit points and is therefore not closed.

1.1.4 Let $A \in \mathbb{R}^{m \times n}$. Show that if $S \subseteq \mathbb{R}^m$ is convex then so is $A^{-1}(S) = \{x \in \mathbb{R}^n : Ax \in S\}$, which is called the preimage of S under the map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Show that the same statement holds if we replace “convex” with “closed”.

Answer

Let $A \in \mathbb{R}^{m \times n}$ and $S \subseteq \mathbb{R}^m$. Define the preimage of S under A as

$$A^{-1}(S) = \{x \in \mathbb{R}^n : Ax \in S\}.$$

Convexity. Assume S is convex. Take any $x, y \in A^{-1}(S)$ and $t \in [0, 1]$. Then $Ax \in S$ and $Ay \in S$. Since A is linear,

$$A(tx + (1 - t)y) = tAx + (1 - t)Ay.$$

Because S is convex, $tAx + (1 - t)Ay \in S$. Hence $A(tx + (1 - t)y) \in S$, which means $tx + (1 - t)y \in A^{-1}(S)$. Thus $A^{-1}(S)$ is convex.

Closedness. Assume S is closed. Let $(x_k)_{k \geq 1} \subseteq A^{-1}(S)$ be a sequence that converges to some $\bar{x} \in \mathbb{R}^n$. For each k , $Ax_k \in S$. The linear map A is continuous (all linear operators on finite-dimensional spaces are continuous), so

$$\lim_{k \rightarrow \infty} Ax_k = A \lim_{k \rightarrow \infty} x_k = A\bar{x}.$$

Since S is closed, the limit of the convergent sequence $(Ax_k) \subseteq S$ must belong to S ; therefore $A\bar{x} \in S$. By definition this means $\bar{x} \in A^{-1}(S)$. Hence $A^{-1}(S)$ contains all its limit points and is closed.

1.1.5 Let $A \in \mathbb{R}^{m \times n}$. Show that if $S \subseteq \mathbb{R}^n$ is convex then so is $A(S) = \{Ax : x \in S\}$, called the image of S under A .

Answer

Let $A \in \mathbb{R}^{m \times n}$ and $S \subseteq \mathbb{R}^n$ be convex. Define the image

$$A(S) = \{Ax : x \in S\}.$$

Take two points $u, v \in A(S)$. By definition, there exist $x, y \in S$ such that $u = Ax$ and $v = Ay$. For any $t \in [0, 1]$, linearity of A gives

$$tu + (1 - t)v = tAx + (1 - t)Ay = A(tx + (1 - t)y).$$

Since S is convex, $tx + (1 - t)y \in S$, so the right-hand side belongs to $A(S)$. Thus $tu + (1 - t)v \in A(S)$, proving $A(S)$ is convex.

1.1.6 Give an example of a matrix $A \in \mathbb{R}^{m \times n}$ and a set $S \subseteq \mathbb{R}^n$ that is closed and convex but such that $A(S)$ is not closed.

Answer

Define

$$S = \{(x, y) \in \mathbb{R}^2 : x > 0, y \geq \frac{1}{x}\}.$$

This is the set of all points lying on or above the curve $y = 1/x$ in the right half-plane.

Why S is convex. The curve $y = 1/x$ for $x > 0$ has second derivative $2/x^3 > 0$, so it is a convex function. The region above its graph (the *epigraph*) is always convex.

Why S is closed. Take any convergent sequence $(x_k, y_k) \in S$ with limit $(\bar{x}, \bar{y})$.

If $\bar{x} = 0$, then $x_k \rightarrow 0$, so $1/x_k \rightarrow \infty$. Since $y_k \geq 1/x_k$, the sequence y_k would have to go to ∞ , contradicting convergence to a finite $\bar{y}$. Hence $\bar{x} \neq 0$.

Therefore $\bar{x} > 0$. Because the inequality $y_k \geq 1/x_k$ is preserved in the limit (the functions are continuous on $(0, \infty)$), we get $\bar{y} \geq 1/\bar{x}$. So $(\bar{x}, \bar{y}) \in S$. Thus S contains all its limit points, i.e., it is closed.

The linear map. Choose $A = \begin{bmatrix} 1 & 0 \end{bmatrix}$, the projection onto the x -axis: $A(x, y) = x$.

The image $A(S)$.

$$A(S) = \{x \in \mathbb{R} : \exists y \text{ with } (x, y) \in S\} = (0, \infty).$$

Indeed, for any $x > 0$ we can take $y = 1/x$ (or any larger value) to get a point in S , so all positive numbers appear. No non-positive number can appear because every point in S has $x > 0$.

Why the image is not closed. The sequence $a_k = 1/k$ ($k = 1, 2, \dots$) lies entirely in $(0, \infty)$, hence in $A(S)$. Its limit is $\lim_{k \rightarrow \infty} a_k = 0$, but $0 \notin (0, \infty) = A(S)$.

Therefore $A(S)$ fails to contain a limit point and is not closed, even though S is closed and convex.

1.2 (4 pts) Polyhedra

1.2.1 Show that if $P \subseteq \mathbb{R}^n$ is a polyhedron, and $A \in \mathbb{R}^{m \times n}$, then $A(P)$ is a polyhedron. Hint: you may use the fact that $P \subseteq \mathbb{R}^{m+n}$ is a polyhedron $\Rightarrow \{x \in \mathbb{R}^n : (x, y) \in P \text{ for some } y \in \mathbb{R}^m\}$ is a polyhedron.

Answer

Let $P \subseteq \mathbb{R}^n$ be a polyhedron, so by definition there exist a matrix $B \in \mathbb{R}^{k \times n}$ and a vector $b \in \mathbb{R}^k$ such that

$$P = \{x \in \mathbb{R}^n : Bx \leq b\}.$$

Let $A \in \mathbb{R}^{m \times n}$ be a linear map. We want to show that

$$A(P) = \{Ax : x \in P\}$$

is also a polyhedron, i.e., can be expressed as $\{y \in \mathbb{R}^m : Cy \leq d\}$ for some C, d .

Step 1: Construct an extended polyhedron in $\mathbb{R}^{n+m}$. Define the set

$$Q = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^m : Bx \leq b, y = Ax\}.$$

The condition $y = Ax$ is a system of m linear equations. Each equation can be replaced by two inequalities:

$$Ax - y \leq 0 \quad \text{and} \quad -(Ax - y) \leq 0,$$

where the inequalities are taken component-wise. Hence Q is equivalently given by

$$Q = \left\{ (x, y) \in \mathbb{R}^{n+m} : \begin{array}{l} Bx \leq b, \\ Ax - y \leq 0, \\ -Ax + y \leq 0 \end{array} \right\}.$$

All constraints are affine inequalities in the joint variable $\begin{pmatrix} x \\ y \end{pmatrix}$. Therefore Q is a polyhedron in $\mathbb{R}^{n+m}$.

Step 2: Relate the projection of Q to $A(P)$. Consider the projection of Q onto its last m coordinates (the y -space):

$$\text{proj}_y(Q) = \{y \in \mathbb{R}^m : \exists x \in \mathbb{R}^n \text{ such that } (x, y) \in Q\}.$$

By the definition of Q , $(x, y) \in Q$ if and only if $x \in P$ and $y = Ax$. Hence

$$\text{proj}_y(Q) = \{y \in \mathbb{R}^m : \exists x \in P \text{ with } y = Ax\} = A(P).$$

Step 3: Apply the projection theorem for polyhedra. The problem statement provides the following fact (which is a consequence of Fourier–Motzkin elimination):

If a set $S \subseteq \mathbb{R}^{n+m}$ is a polyhedron, then its projection onto any subset of coordinates is also a polyhedron.

Since $Q \subseteq \mathbb{R}^{n+m}$ is a polyhedron, its projection $\text{proj}_y(Q) = A(P)$ must be a polyhedron. Thus $A(P)$ is indeed a polyhedron.

1.2.2 Show that if $Q \subseteq \mathbb{R}^m$ is a polyhedron, and $A \in \mathbb{R}^{m \times n}$, then $A^{-1}(Q)$ is a polyhedron.

Answer

Let $Q \subseteq \mathbb{R}^m$ be a polyhedron. By definition, there exist a matrix $D \in \mathbb{R}^{k \times m}$ and a vector $e \in \mathbb{R}^k$ such that

$$Q = \{y \in \mathbb{R}^m : Dy \leq e\}.$$

Let $A \in \mathbb{R}^{m \times n}$ be a matrix. The preimage of Q under A is

$$A^{-1}(Q) = \{x \in \mathbb{R}^n : Ax \in Q\}.$$

Substituting the description of Q gives

$$A^{-1}(Q) = \{x \in \mathbb{R}^n : D(Ax) \leq e\} = \{x \in \mathbb{R}^n : (DA)x \leq e\}.$$

The matrix $C = DA \in \mathbb{R}^{k \times n}$ defines a finite system of linear inequalities in x . Therefore $A^{-1}(Q)$ is a polyhedron in $\mathbb{R}^n$.

2 (14 pts) – Convex functions

2.1 (2 pts)

Prove that the entropy function, defined as

$$f(x) = -\sum_{i=1}^n x_i \log(x_i),$$

with $\text{dom}(f) = \{x \in \mathbb{R}_+^n : \sum_{i=1}^n x_i = 1\}$, is strictly concave.

Answer

The function is

$$f(x) = -\sum_{i=1}^n x_i \log(x_i), \quad \text{dom}(f) = \left\{x \in \mathbb{R}_{++}^n : \sum_{i=1}^n x_i = 1\right\}.$$

Write f as a sum of coordinate functions:

$$f(x) = \sum_{i=1}^n h(x_i), \quad \text{where } h(t) = -t \log t, \quad t > 0.$$

For $t > 0$ we compute the second derivative:

$$h'(t) = -\log t - 1, \quad h''(t) = -\frac{1}{t} < 0.$$

Since $h''(t) < 0$ on $(0, \infty)$, the function h is strictly concave.

A sum of strictly concave functions is strictly concave: for any $x \neq y$ in the domain there is at least one coordinate i with $x_i \neq y_i$, giving a strict inequality in that term, while the remaining terms satisfy the non-strict concavity inequality. The domain $\text{dom}(f)$ is a convex subset of $\mathbb{R}_{++}^n$, so the restriction of a strictly concave function to a convex set remains strictly concave. Hence f is strictly concave on its domain.

2.2 (4 pts)

Let f be twice differentiable, with $\text{dom}(f)$ convex. Prove that f is convex if and only if

$$(\nabla f(x) - \nabla f(y))^T(x - y) \geq 0,$$

for all x, y . This property is called monotonicity of the gradient ∇f .

Answer

Let f be twice differentiable with $\text{dom}(f)$ convex. We prove that

$$f \text{ is convex} \iff (\nabla f(x) - \nabla f(y))^T(x - y) \geq 0 \quad \text{for all } x, y \in \text{dom}(f).$$

$\Rightarrow$ **(Convexity implies monotone gradient)**. Assume f is convex. For any x, y , the first-order characterization of convexity gives

$$f(y) \geq f(x) + \nabla f(x)^T(y - x), \quad f(x) \geq f(y) + \nabla f(y)^T(x - y).$$

Add the two inequalities:

$$f(x) + f(y) \geq f(x) + f(y) + \nabla f(x)^T(y - x) + \nabla f(y)^T(x - y).$$

Cancel $f(x) + f(y)$ from both sides and obtain

$$0 \geq \nabla f(x)^T(y - x) + \nabla f(y)^T(x - y) = (\nabla f(x) - \nabla f(y))^T(y - x).$$

Multiplying by -1 yields

$$(\nabla f(x) - \nabla f(y))^T(x - y) \geq 0,$$

which is the desired monotonicity.

$\Leftarrow$ **(Monotone gradient implies convexity)**. Assume $(\nabla f(u) - \nabla f(v))^T(u - v) \geq 0$ for all u, v . Fix x, y and consider the function $\varphi(t) = f(x + t(y - x))$ for $t \in [0, 1]$. Since f is differentiable, φ is differentiable and

$$\varphi'(t) = \nabla f(x + t(y - x))^T(y - x).$$

Apply the monotonicity condition to $u = x + t(y - x)$ and $v = x$ (for $t > 0$):

$$(\nabla f(x + t(y - x)) - \nabla f(x))^T(t(y - x)) \geq 0.$$

Divide by $t > 0$ (for $t = 0$ the inequality is trivial):

$$\nabla f(x + t(y - x))^T(y - x) \geq \nabla f(x)^T(y - x).$$

Thus $\varphi'(t) \geq \nabla f(x)^T(y - x)$ for all $t \in (0, 1]$. Integrate from 0 to 1:

$$f(y) - f(x) = \varphi(1) - \varphi(0) = \int_0^1 \varphi'(t) dt \geq \int_0^1 \nabla f(x)^T(y - x) dt = \nabla f(x)^T(y - x).$$

We have obtained the first-order convexity inequality

$$f(y) \geq f(x) + \nabla f(x)^T(y - x) \quad \text{for all } x, y,$$

which is well known to be equivalent to convexity of f (and follows by applying the inequality to convex combinations). Hence f is convex.

2.3 (2 pts)

Give an example of a strictly convex function that does not attain its infimum.

Answer

Consider $f(x) = e^x$, defined on $\mathbb{R}$.

$f''(x) = e^x > 0$ for all $x \in \mathbb{R}$, hence f is strictly convex.

$\inf_{x \in \mathbb{R}} f(x) = \lim_{x \rightarrow -\infty} e^x = 0$, but $f(x) > 0$ for every x . Therefore the infimum exists (it is 0) but is never attained.

2.4 (3 pts)

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be coercive provided that $f(x) \rightarrow \infty$ as $\|x\|_2 \rightarrow \infty$. A key fact about coercive functions is that they attain their infimums. Prove that a twice differentiable, strongly convex function is coercive and hence attains its infimum. Hint: use Q3 part (b.iv).

Answer

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable and strongly convex with constant $m > 0$. By the strong convexity inequality (Problem 4(b)(iv)), for all $x, y \in \mathbb{R}^n$ we have

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|_2^2.$$

Fix an arbitrary point $x_0 \in \mathbb{R}^n$. Then for any y ,

$$f(y) \geq f(x_0) + \nabla f(x_0)^\top (y - x_0) + \frac{m}{2} \|y - x_0\|_2^2.$$

By the Cauchy–Schwarz inequality,

$$|\nabla f(x_0)^\top (y - x_0)| \leq \|\nabla f(x_0)\|_2 \|y - x_0\|_2.$$

Hence

$$f(y) \geq f(x_0) - \|\nabla f(x_0)\|_2 \|y - x_0\|_2 + \frac{m}{2} \|y - x_0\|_2^2.$$

Consider the behavior as $\|y\|_2 \rightarrow \infty$. Then $\|y - x_0\|_2 \rightarrow \infty$ as well. The right-hand side is a quadratic in $\|y - x_0\|_2$ with positive leading coefficient $m/2$, so it tends to $+\infty$. Thus $\lim_{\|y\|_2 \rightarrow \infty} f(y) = \infty$; i.e., f is **coercive**, thus attains its infimum.

2.5 (3 pts)

Prove that the maximum of a convex function over a bounded polyhedron must occur at one of the vertices. Hint: you may use the fact that a bounded polyhedron can be represented as the convex hull of its vertices.

Answer

Let $P \subseteq \mathbb{R}^n$ be a bounded polyhedron and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex. Since P is bounded, it is the convex hull of its finite set of vertices (extreme points), say $v_1, \dots, v_k$.

For any $x \in P$, there exist $\lambda_i \geq 0$ with $\sum_{i=1}^k \lambda_i = 1$ such that

$$x = \sum_{i=1}^k \lambda_i v_i.$$

By convexity of f ,

$$f(x) = f\left(\sum_{i=1}^k \lambda_i v_i\right) \leq \sum_{i=1}^k \lambda_i f(v_i).$$

The right-hand side is a weighted average of the numbers $f(v_i)$; it cannot exceed their maximum:

$$\sum_{i=1}^k \lambda_i f(v_i) \leq \max_{1 \leq i \leq k} f(v_i).$$

Thus for every $x \in P$,

$$f(x) \leq \max_i f(v_i).$$

On the other hand, the vertices themselves are elements of P , and so comparing maximizing over the entire set and some points (vertices) of the set results in

$$\max_{x \in P} f(x) \geq \max_i f(v_i).$$

Combining the two inequalities yields

$$\max_{x \in P} f(x) = \max_i f(v_i).$$

Since the right-hand side is the value of f at (at least) one of the vertices, the maximum of f over P is attained at a vertex.

3 (10 pts (bonus)) – Partial optimization with ℓ_2 penalties

Consider the problem

$$\min_{\beta, \sigma \geq 0} f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n g(\beta_i, \sigma_i), \quad (1)$$

for some convex f with domain $\mathbb{R}^n$, $\lambda \geq 0$, and

$$g(x, y) = \begin{cases} \frac{x^2}{y} + y, & y > 0, \\ 0, & (x, y) = (0, 0), \\ +\infty, & \text{otherwise,} \end{cases}$$

In other words, the problem (1) is just the weighted ℓ_2 penalized problem

$$\min_{\beta, \sigma \geq 0} f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n \left(\frac{\beta_i^2}{\sigma_i} + \sigma_i \right),$$

but being careful to treat the i th term in the sum as zero when $\beta_i = \sigma_i = 0$.

3.1 (5 pts)

Prove that g is convex. Hence argue that (1) is a convex problem. Note that this means we can perform partial optimization in (1) and expect it to return another convex problem. Hint: use the definition of convexity.

Answer

The function $g : \mathbb{R}^2 \rightarrow \mathbb{R} \cup \{+\infty\}$ is defined as

$$g(x, y) = \begin{cases} \frac{x^2}{y} + y, & y > 0, \\ 0, & (x, y) = (0, 0), \\ +\infty, & \text{otherwise.} \end{cases}$$

Its effective domain is $\text{dom}(g) = \{(x, y) : y > 0\} \cup \{(0, 0)\}$.

Step 1: Convexity on the interior $y > 0$. For $y > 0$, g is twice differentiable. Its gradient and Hessian are

$$\nabla g(x, y) = \begin{pmatrix} \frac{2x}{y} \\ -\frac{x^2}{y^2} + 1 \end{pmatrix}, \quad \nabla^2 g(x, y) = \begin{pmatrix} \frac{2}{y} & -\frac{2x}{y^2} \\ -\frac{2x}{y^2} & \frac{2x^2}{y^3} \end{pmatrix}.$$

To check positive semidefiniteness of the Hessian, we compute its leading principal minors. The first entry is $\frac{2}{y} > 0$. The determinant is

$$\det \nabla^2 g = \frac{2}{y} \cdot \frac{2x^2}{y^3} - \left(-\frac{2x}{y^2}\right)^2 = \frac{4x^2}{y^4} - \frac{4x^2}{y^4} = 0 \geq 0.$$

Thus both leading principal minors are non-negative; together with $\frac{2}{y} > 0$, the Hessian is positive semidefinite for every (x, y) with $y > 0$. Hence g is convex on the open half-plane $\{y > 0\}$.

Step 2: Convexity at the boundary point $(0, 0)$. Take any (x, y) with $y > 0$ and any $\theta \in (0, 1)$. The convex combination of $(0, 0)$ and (x, y) is

$$(x_\theta, y_\theta) = \theta(0, 0) + (1 - \theta)(x, y) = ((1 - \theta)x, (1 - \theta)y),$$

which still satisfies $y_\theta > 0$. We compute directly:

$$\begin{aligned} g(x_\theta, y_\theta) &= \frac{((1 - \theta)x)^2}{(1 - \theta)y} + (1 - \theta)y \\ &= (1 - \theta) \frac{x^2}{y} + (1 - \theta)y \\ &= (1 - \theta) g(x, y) + \theta \cdot 0 \\ &= \theta g(0, 0) + (1 - \theta) g(x, y). \end{aligned}$$

Equality holds, so the convexity inequality is satisfied. The case where both points are $(0, 0)$ is trivial. Together with Step 1, g is convex on its entire effective domain (and hence as an extended-value function on $\mathbb{R}^2$).

Step 3: Convexity of problem (1). The optimization problem is

$$\min_{\beta, \sigma \geq 0} f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n g(\beta_i, \sigma_i).$$

Here f is convex on $\mathbb{R}^n$, $\lambda \geq 0$, and each $g(\beta_i, \sigma_i)$ is convex by the argument above. A non-negative weighted sum of convex functions is convex, so the objective is convex in (β, σ) . The feasible set $\{\sigma \geq 0\}$ is an intersection of half-spaces, hence convex. Thus (1) is a convex optimization problem.

3.2 (2 pts)

Argue that $\min_{y \geq 0} g(x, y) = 2|x|$.

Answer

We prove $\min_{y \geq 0} g(x, y) = 2|x|$ for every $x \in \mathbb{R}$.

Case $x = 0$. If $y > 0$, $g(0, y) = 0^2/y + y = y$, whose infimum as $y \rightarrow 0^+$ is 0 (not attained). At $y = 0$, the only admissible point is $(0, 0)$ with $g(0, 0) = 0$. Hence $\min_{y \geq 0} g(0, y) = 0 = 2|0|$.

Case $x \neq 0$. For $y = 0$, the point $(x, 0)$ with $x \neq 0$ is not in the effective domain, so $g(x, 0) = +\infty$ and cannot be the minimum. We therefore restrict to $y > 0$. Consider the function $h(y) = \frac{x^2}{y} + y$ for $y > 0$. h is convex, and its derivative is $h'(y) = -\frac{x^2}{y^2} + 1$. Setting $h'(y) = 0$ gives $y^2 = x^2$, so $y = |x|$ (since $y > 0$). Second derivative $h''(y) = \frac{2x^2}{y^3} > 0$

confirms a minimum. The minimum value is $h(|x|) = \frac{x^2}{|x|} + |x| = 2|x|$.

Both cases yield the same expression. Therefore $\min_{y \geq 0} g(x, y) = 2|x|$ for all x .

3.3 (3 pts)

Argue that minimizing over $\sigma \geq 0$ in (1) gives the ℓ_1 penalized problem

$$\min_{\beta} f(\beta) + \lambda \|\beta\|_1.$$

Answer

Consider the original problem

$$\min_{\beta, \sigma \geq 0} f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n g(\beta_i, \sigma_i).$$

For a fixed β , the minimization over $\sigma \geq 0$ can be carried out separately for each coordinate because the sum is separable:

$$\begin{aligned} \inf_{\sigma \geq 0} \left[f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n g(\beta_i, \sigma_i) \right] \\ = f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n \left(\min_{\sigma_i \geq 0} g(\beta_i, \sigma_i) \right). \end{aligned}$$

From part (b), $\min_{\sigma \geq 0} g(\beta_i, \sigma) = 2|\beta_i|$ for each i . Substituting this into the expression above yields

$$\min_{\sigma \geq 0} \left[f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n g(\beta_i, \sigma_i) \right] = f(\beta) + \frac{\lambda}{2} \sum_{i=1}^n 2|\beta_i| = f(\beta) + \lambda \|\beta\|_1.$$

Finally, minimizing over β gives the ℓ_1 penalized problem

$$\min_{\beta} f(\beta) + \lambda \|\beta\|_1,$$

which is exactly what we wanted to show.

4 (18 pts) – Lipschitz gradients and strong convexity

Let f be convex and twice continuously differentiable.

4.1 (10 pts) Lipschitz gradients

Show the equivalence of the following statements.

Answer

Assume f is convex and twice continuously differentiable on $\mathbb{R}^n$.

(i) $\Rightarrow$ (ii). Suppose ∇f is L -Lipschitz, i.e.

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2 \quad \forall x, y.$$

By the Cauchy–Schwarz inequality,

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \leq \|\nabla f(x) - \nabla f(y)\|_2 \|x - y\|_2.$$

Combining the two gives

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \leq L\|x - y\|_2^2.$$

(ii) $\Rightarrow$ (iii). Assume $(\nabla f(x) - \nabla f(y))^\top (x - y) \leq L\|x - y\|_2^2$ for all x, y . Fix $x \in \mathbb{R}^n$ and an arbitrary direction $v \in \mathbb{R}^n$. For $t > 0$ set $y = x + tv$. Then the inequality becomes

$$(\nabla f(x + tv) - \nabla f(x))^\top (tv) \leq L\|tv\|_2^2 = Lt^2\|v\|_2^2.$$

Divide both sides by t^2 :

$$\left(\frac{\nabla f(x + tv) - \nabla f(x)}{t} \right)^\top v \leq L\|v\|_2^2.$$

Taking the limit $t \rightarrow 0^+$ and using the definition of the directional derivative of ∇f ,

$$\lim_{t \rightarrow 0^+} \frac{\nabla f(x + tv) - \nabla f(x)}{t} = \nabla^2 f(x) v,$$

we obtain

$$v^\top \nabla^2 f(x) v \leq L\|v\|_2^2.$$

Since v is arbitrary, this means $v^\top (\nabla^2 f(x) - LI)v \leq 0$ for all v , i.e.

$$\nabla^2 f(x) \preceq LI \quad \forall x.$$

(iii) $\Rightarrow$ (iv). Assume $\nabla^2 f(x) \preceq LI$ for all x . Fix x, y and define $\varphi(t) = f(x + t(y - x))$ for $t \in [0, 1]$. φ is twice differentiable with

$$\varphi'(t) = \nabla f(x + t(y - x))^\top (y - x), \quad \varphi''(t) = (y - x)^\top \nabla^2 f(x + t(y - x))(y - x).$$

Using the Hessian bound,

$$\varphi''(t) \leq L\|y - x\|_2^2.$$

Now apply the second-order Taylor expansion (or integral form of the remainder):

$$\varphi(1) = \varphi(0) + \varphi'(0) + \int_0^1 (1 - t) \varphi''(t) dt.$$

Since $\varphi(1) = f(y)$, $\varphi(0) = f(x)$, $\varphi'(0) = \nabla f(x)^\top (y - x)$, substituting the bound on φ'' yields

$$\begin{aligned} f(y) &\leq f(x) + \nabla f(x)^\top (y - x) + \int_0^1 (1 - t) L\|y - x\|_2^2 dt \\ &= f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|_2^2. \end{aligned}$$

(iv) $\Rightarrow$ (ii). Assume the quadratic upper bound holds for all x, y :

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2}\|y - x\|_2^2.$$

Write the same inequality with the roles of x and y swapped:

$$f(x) \leq f(y) + \nabla f(y)^\top (x - y) + \frac{L}{2}\|x - y\|_2^2.$$

Adding the two inequalities gives

$$f(x) + f(y) \leq f(x) + f(y) + \nabla f(x)^\top (y - x) + \nabla f(y)^\top (x - y) + L\|x - y\|_2^2.$$

Cancel $f(x) + f(y)$ and rearrange:

$$0 \leq (\nabla f(y) - \nabla f(x))^\top (x - y) + L\|x - y\|_2^2,$$

which is equivalent to

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \leq L\|x - y\|_2^2.$$

(iii) $\Rightarrow$ (i). Assume $\nabla^2 f(x) \preceq LI$ for all x . Since f is convex, we also have $\nabla^2 f(x) \succeq 0$, so all eigenvalues of $\nabla^2 f(x)$ lie in $[0, L]$. For a symmetric positive semidefinite matrix, the operator norm (spectral norm) equals the largest eigenvalue; hence

$$\|\nabla^2 f(x)\|_{\text{op}} \leq L \quad \forall x.$$

Now represent the gradient difference using the fundamental theorem of calculus:

$$\nabla f(y) - \nabla f(x) = \int_0^1 \frac{d}{dt} \nabla f(x + t(y - x)) dt = \left(\int_0^1 \nabla^2 f(x + t(y - x)) dt \right) (y - x).$$

Let $M = \int_0^1 \nabla^2 f(x + t(y - x)) dt$. Then $\nabla f(y) - \nabla f(x) = M(y - x)$.

Take norms:

$$\|\nabla f(y) - \nabla f(x)\|_2 \leq \|M\|_{\text{op}} \|y - x\|_2.$$

We bound $\|M\|_{\text{op}}$ using the triangle inequality for integrals (the norm of an integral is at most the integral of the norm):

$$\|M\|_{\text{op}} = \left\| \int_0^1 \nabla^2 f(x + t(y - x)) dt \right\|_{\text{op}} \leq \int_0^1 \|\nabla^2 f(x + t(y - x))\|_{\text{op}} dt.$$

Since each integrand has operator norm $\leq L$,

$$\|M\|_{\text{op}} \leq \int_0^1 L dt = L.$$

Therefore

$$\|\nabla f(y) - \nabla f(x)\|_2 \leq L\|y - x\|_2,$$

i.e., ∇f is L -Lipschitz.

4.2 (8 pts) Strong convexity

Show the equivalence of the following statements.

Answer

Assume f is convex and twice continuously differentiable on $\mathbb{R}^n$, and let $m > 0$.

(i) $\Rightarrow$ (ii). f is strongly convex with constant m , meaning $g(x) = f(x) - \frac{m}{2}\|x\|_2^2$ is convex. The gradient of g is $\nabla g(x) = \nabla f(x) - mx$. Convexity of g gives the monotone gradient inequality:

$$(\nabla g(x) - \nabla g(y))^\top (x - y) \geq 0 \quad \forall x, y.$$

Substituting the expression for ∇g :

$$(\nabla f(x) - mx - \nabla f(y) + my)^\top (x - y) \geq 0,$$

which simplifies to

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq m\|x - y\|_2^2.$$

(ii) $\Rightarrow$ (iii). Assume $(\nabla f(x) - \nabla f(y))^\top (x - y) \geq m\|x - y\|_2^2$ for all x, y . Fix x and an arbitrary direction v . For $t > 0$ set $y = x + tv$. Then

$$(\nabla f(x + tv) - \nabla f(x))^\top (tv) \geq m\|tv\|_2^2 = mt^2\|v\|_2^2.$$

Divide by t^2 :

$$\left(\frac{\nabla f(x + tv) - \nabla f(x)}{t} \right)^\top v \geq m\|v\|_2^2.$$

Taking the limit $t \rightarrow 0^+$ yields

$$v^\top \nabla^2 f(x) v \geq m\|v\|_2^2.$$

Since v is arbitrary, this is equivalent to

$$\nabla^2 f(x) \succeq mI \quad \forall x.$$

(iii) $\Rightarrow$ (iv). Assume $\nabla^2 f(x) \succeq mI$ for all x . Fix x, y and define $\varphi(t) = f(x + t(y - x))$ for $t \in [0, 1]$. Then

$$\varphi'(t) = \nabla f(x + t(y - x))^\top (y - x), \quad \varphi''(t) = (y - x)^\top \nabla^2 f(x + t(y - x)) (y - x) \geq m \|y - x\|_2^2.$$

Using the second-order Taylor expansion with integral remainder,

$$\varphi(1) = \varphi(0) + \varphi'(0) + \int_0^1 (1 - t) \varphi''(t) dt.$$

Substituting the lower bound for φ'' gives

$$\begin{aligned} f(y) &= f(x) + \nabla f(x)^\top (y - x) + \int_0^1 (1 - t) \varphi''(t) dt \\ &\geq f(x) + \nabla f(x)^\top (y - x) + \int_0^1 (1 - t) m \|y - x\|_2^2 dt \\ &= f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|_2^2. \end{aligned}$$

(iv) $\Rightarrow$ (i). Assume $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|_2^2$ for all x, y . Define $g(x) = f(x) - \frac{m}{2} \|x\|_2^2$. Its gradient is $\nabla g(x) = \nabla f(x) - mx$. We show g is convex by establishing the first-order inequality $g(y) \geq g(x) + \nabla g(x)^\top (y - x)$.

Start from the assumed inequality for $f(y)$:

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|_2^2.$$

Subtract $\frac{m}{2} \|y\|_2^2$ from both sides:

$$g(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|_2^2 - \frac{m}{2} \|y\|_2^2.$$

Expand $\|y - x\|_2^2 = \|y\|_2^2 - 2y^\top x + \|x\|_2^2$. Then

$$\frac{m}{2} \|y - x\|_2^2 - \frac{m}{2} \|y\|_2^2 = -m y^\top x + \frac{m}{2} \|x\|_2^2.$$

Thus

$$g(y) \geq f(x) + \nabla f(x)^\top (y - x) - m y^\top x + \frac{m}{2} \|x\|_2^2.$$

Now write $f(x) = g(x) + \frac{m}{2} \|x\|_2^2$ and $\nabla f(x) = \nabla g(x) + mx$. Substituting,

$$\begin{aligned} g(y) &\geq g(x) + \frac{m}{2} \|x\|_2^2 + (\nabla g(x) + mx)^\top (y - x) - m y^\top x + \frac{m}{2} \|x\|_2^2 \\ &= g(x) + \nabla g(x)^\top (y - x) + mx^\top (y - x) - m y^\top x + m \|x\|_2^2 \\ &= g(x) + \nabla g(x)^\top (y - x) + m(x^\top y - \|x\|_2^2 - y^\top x + \|x\|_2^2) \\ &= g(x) + \nabla g(x)^\top (y - x). \end{aligned}$$

Hence $g(y) \geq g(x) + \nabla g(x)^\top (y - x)$, which is the first-order characterization of convexity for g . Therefore g is convex, i.e., f is strongly convex with constant m .

5 (18 pts) – Solving optimization problems with CVX

5.1 (10 pts) SVM

- i. Load the training data. Using CVX, solve the SVM problem with $C = 1$. Report the optimal criterion value, and the optimal coefficients $\beta \in \mathbb{R}^2$ and intercept $\beta_0 \in \mathbb{R}$.

Answer

Optimal criterion value: **36.748932649263416**

$\beta = [1.41967191, 1.24607477]$

$\beta_0 = -2.823727594195054$

- ii. Plot the training data and the decision boundary.

Answer

The decision boundary is shown in Figure 1.

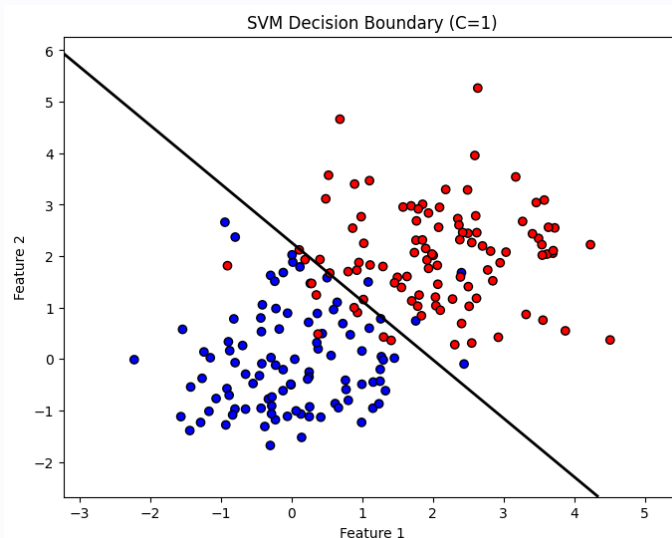


Figure 1: SVM decision boundary for $C = 1$.

- iii. Solve the dual-like problem and compare.

Answer

Solving the dual-like problem:

Dual optimal value: 36.54676673663892

$\tilde{X}^\top w = [1.41922771, 1.24558068]$

The primal optimal value was 36.748932649263416.

The values match as expected (small numerical differences due to solver accuracy).

- iv. Varying $C = 2^a$, $a = -5, \dots, 5$. Plot misclassification error versus C .

Answer

The plot is shown in Figure 2.

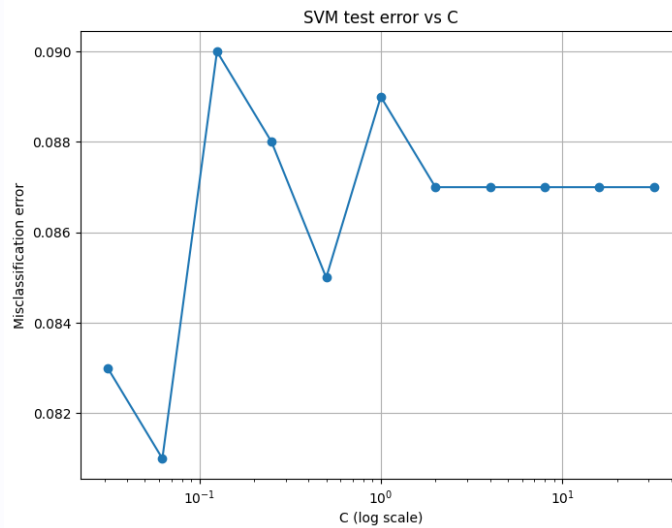


Figure 2: Test misclassification error vs C (log scale).

5.2 (8 pts) DCP expressions

For each set of expressions, briefly explain why each defines a convex set, then give an equivalent DCP expression.

Answer

- i. $\|(x, y, z)\|_2^2 \leq 1$ **Convexity:** The left side is `square(x) + square(y) + square(z)`, a sum of convex functions. A convex function $\leq$ constant is a convex set. **DCP:** `square(x) + square(y) + square(z) <= 1`
- ii. $\sqrt{x^2 + 1} \leq 3x + y$ **Convexity:** Left side is `norm2(x, 1)`, convex. Constraint convex $\leq$ affine is allowed. **DCP:** `norm2(x, 1) <= 3*x + y`
- iii. $1/x + 2/y \leq 5, x > 0, y > 0$ **Convexity:** `inv_pos(x)` and `inv_pos(y)` are convex. Sum of convex functions $\leq$ constant is convex. **DCP:** `inv_pos(x) + 2*inv_pos(y) <= 5`
- iv. $(x + y)^2 / \sqrt{y} \leq x - y + 5, y > 0$ **Convexity:** `quad_over_lin(x + y, sqrt(y))` is convex. Right side is affine $\rightarrow$ convex $\leq$ affine is convex. **DCP:** `quad_over_lin(x + y, sqrt(y)) <= x - y + 5`
- v. $(x + z)y \geq 1, x + z \geq 0, y \geq 0$ **Convexity:** This is a rotated second-order cone: $\sqrt{(x + z - y)^2 + 4} \leq x + z + y$. Left side is `norm2(x + z - y, 2)`, convex; right side is affine. **DCP:** `norm2(x + z - y, 2) <= x + z + y`
- vi. $\|(x + 2y, x - y)\|_2 = 0$ **Convexity:** A norm equals zero iff its argument is the zero vector, which is an affine subspace (convex). **DCP:** `x + 2*y == 0; x - y == 0`

- vii. $x\sqrt{y} \geq 1, x \geq 0, y \geq 0$ **Convexity:** The inequality is $x \geq 1/\sqrt{y}$. `inv_pos(sqrt(y))` is convex, so $\text{convex} \leq \text{affine}$ gives a convex set. **DCP:** `inv_pos(sqrt(y)) <= x`
- viii. $\log(e^{y-1} + e^{x/2}) \leq -e^x$ **Convexity:** Left side is `log_sum_exp(y - 1, x/2)`, convex; `exp(x)` is convex. Sum of convex functions ≤ 0 is convex. **DCP:** `log_sum_exp(y - 1, x/2) + exp(x) <= 0`

APPENDIX: P5.A CODE

Listing 1: Full SVM implementation.

```

1 import numpy as np
2 import cvxpy as cp
3 import matplotlib.pyplot as plt
4
5 train_data = np.loadtxt('xy_train.csv', delimiter=',')
6 X_train = train_data[:, :2]
7 y_train = train_data[:, 2]
8
9 test_data = np.loadtxt('xy_test.csv', delimiter=',')
10 X_test = test_data[:, :2]
11 y_test = test_data[:, 2]
12
13 if np.any(y_train == 0):
14     y_train = 2 * y_train - 1
15     y_test = 2 * y_test - 1
16
17 # --- Part i: SVM with C=1 ---
18 n_train, p = X_train.shape
19 C = 1.0
20 beta = cp.Variable(p)
21 beta0 = cp.Variable()
22 xi = cp.Variable(n_train, nonneg=True)
23
24 objective = 0.5 * cp.sum_squares(beta) + C * cp.sum(xi)
25 constraints = [cp.multiply(y_train, X_train @ beta + beta0) >= 1 - xi
26               ↪ ]
27
28 prob = cp.Problem(cp.Minimize(objective), constraints)
29 prob.solve()
30
31 print("Optimal value:", prob.value)
32 print("beta:", beta.value)
33 print("beta0:", beta0.value)
34
35 # --- Part ii: Plot decision boundary ---
36 x_min, x_max = X_train[:, 0].min() - 1, X_train[:, 0].max() + 1
37 y_min, y_max = X_train[:, 1].min() - 1, X_train[:, 1].max() + 1

```

```

37 xx, yy = np.meshgrid(np.linspace(x_min, x_max, 200),
38                       np.linspace(y_min, y_max, 200))
39 z = beta0.value + beta.value[0] * xx + beta.value[1] * yy
40
41 plt.figure(figsize=(8, 6))
42 plt.contour(xx, yy, z, levels=[0], colors='k', linewidths=2)
43 plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train, cmap='bwr',
44             ↪ edgecolors='k')
45 plt.xlabel('Feature 1')
46 plt.ylabel('Feature 2')
47 plt.title('SVM Decision Boundary (C=1)')
48 plt.show()
49
50 # --- Part iii: Dual-like problem ---
51 X_tilde = y_train[:, None] * X_train
52 w = cp.Variable(n_train)
53 obj_dual = -0.5 * cp.sum_squares(X_tilde.T @ w) + cp.sum(w)
54 constraints_dual = [w >= 0, w <= C, w @ y_train == 0]
55 prob_dual = cp.Problem(cp.Maximize(obj_dual), constraints_dual)
56 prob_dual.solve()
57
58 print("Dual optimal value:", prob_dual.value)
59 print("beta from dual (X_tilde^T w):", (X_tilde.T @ w).value)
60
61 # --- Part iv: Varying C and test error ---
62 C_values = [2**a for a in range(-5, 6)]
63 errors = []
64
65 for C in C_values:
66     beta_c = cp.Variable(p)
67     beta0_c = cp.Variable()
68     xi_c = cp.Variable(n_train, nonneg=True)
69
70     obj_c = 0.5 * cp.sum_squares(beta_c) + C * cp.sum(xi_c)
71     constraints_c = [cp.multiply(y_train, X_train @ beta_c + beta0_c)
72                     ↪ >= 1 - xi_c]
73
74     prob_c = cp.Problem(cp.Minimize(obj_c), constraints_c)
75     prob_c.solve()
76
77     scores = beta0_c.value + X_test @ beta_c.value
78     y_pred = np.sign(scores)
79     error = np.mean(y_pred != y_test)
80     errors.append(error)
81
82 plt.figure(figsize=(8, 6))
83 plt.semilogx(C_values, errors, marker='o')
84 plt.xlabel('C (log scale)')
85 plt.ylabel('Misclassification error')
86 plt.title('SVM test error vs C')
87 plt.grid(True)
88 plt.show()

```